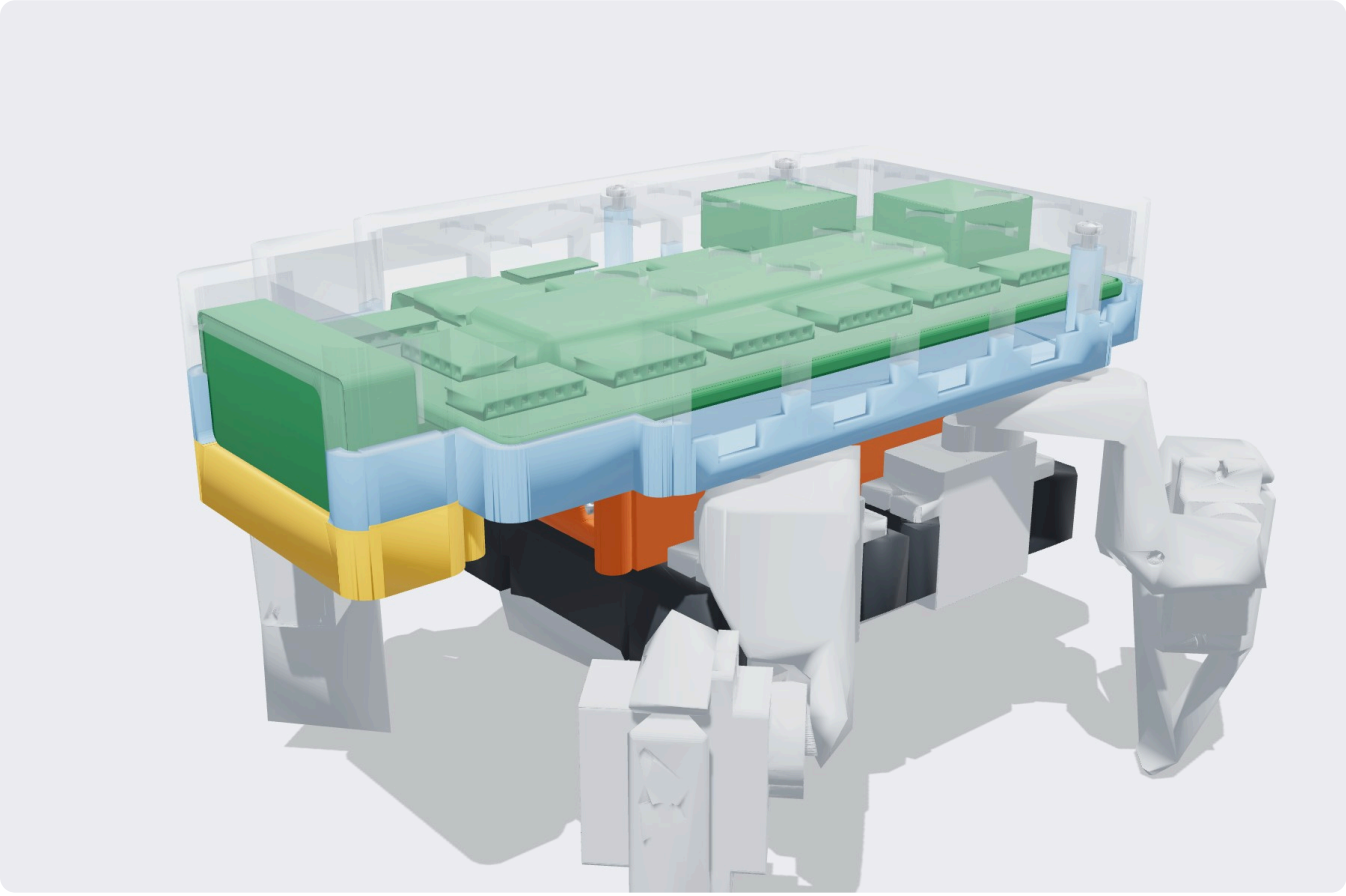
ASSEMBLY GUIDE · V6.3 · 12 NOVEMBER 2026

Atech-crab assembly

Sesame, Dorian Todd's open-source quadruped, with an Atech 14-port board as its brain. Sesame's frame, servos and legs stay. Its top cover is replaced by a three-part deck that snaps together and screws down, the board sits on it, and a lid closes it.



IN THIS GUIDE

What you need	2
1–2 · The frame and the spine	3
3–4 · The head, the plate snaps on	4
5–6 · Screws, board, lid	5
Wiring and software	6

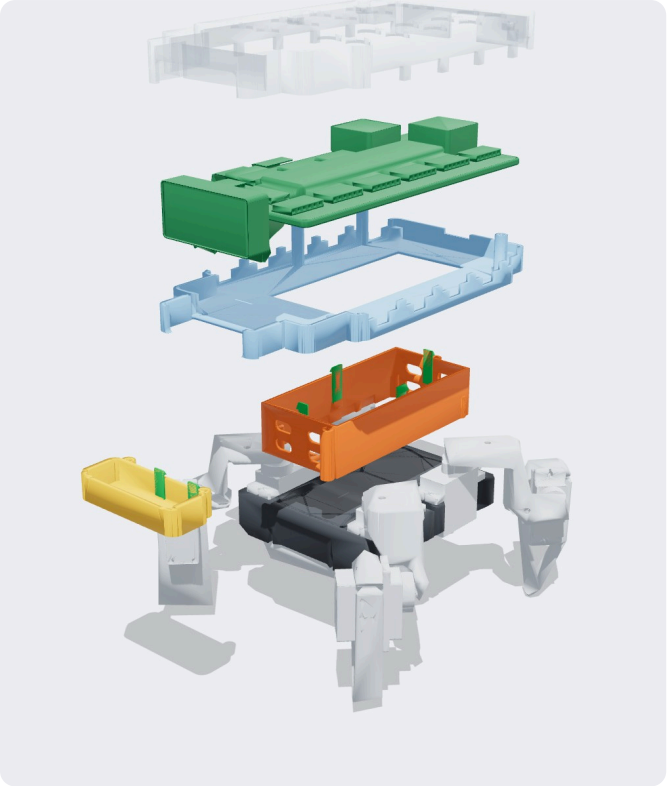
Watch it first: the video `atech_crab_v6.3_assembly.mp4` (50 s) and the animated 3D assembly `docs/assembly/index.html` play every step.

STATUS

The deck, lid and frame pass the fit check in CAD: the snap hooks, every screw's thread, the wire bobbins. **Not yet printed or assembled.** The servo driver runs on the emulated board but is **not yet tried with MG90S servos on a real board.**

Print the deck and test-fit it on the frame, with the servos in, before you print more than one set.

What you need



Deck part 2a, the **spine**	screws to the frame
Deck part 2b, the **head**	holds the screen
Deck part 1, the **plate**	snaps on, carries the board
Snap hooks	4 on the spine, 2 on the head
Atech 14-port board	on the plate's four posts
The **lid**	3 screws, flush heads

PRINTED PARTS (14)

Sesame's internal frame (with a lead slot per servo), the bottom cover, femurs L1 L2 R1 R2, feet L3 L4 R3 R4, deck parts 1, 2a and 2b, and the lid. All are in `atech/print/stl/`, the right way up; sliced files for Bambu H2S, H2D, X2D and P2S are in `atech/print/gcode/`.

WIRE PARTS (16)

4 wire bobbins, 4 port jumper holders, 8 servo plug sleeves, in `atech/print/stl_wires/`.

HARDWARE

8 × MG90S servos (metal gear, 180°) · Atech 14-port board · Atech USB-C module · Atech motion sensor module · Atech screen module with its cable · male-to-male jumper wires · a 5 V 3 A supply for the servos.

SCREWS: TWO KINDS

SCREW	FOR	BUY
M2 × 6 mm pan head, self-tapping, head Ø 4.5 mm	the whole body	about 55
M2.5 × 5 mm machine screw	the 8 servo horns only (they thread into the metal shafts)	8

The deck and lid use 15 of the M2 × 6 screws (step 2: 6 + 2 for the bottom cover; step 5: 4; step 6: 3). The rest go into Sesame's base, as in its build guide.

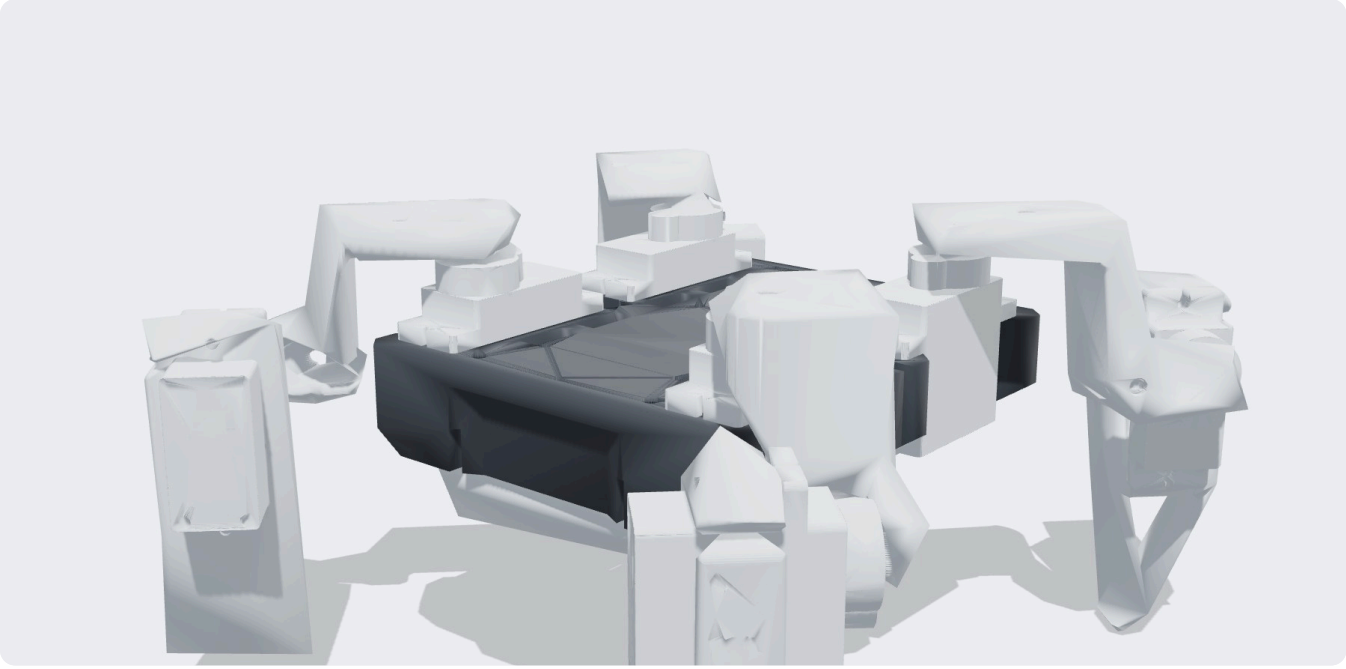
TOOLS

A PH0 screwdriver. A multimeter, to find pin 1 on the ports before wiring (page 6).

1 The base, and a slot for every lead

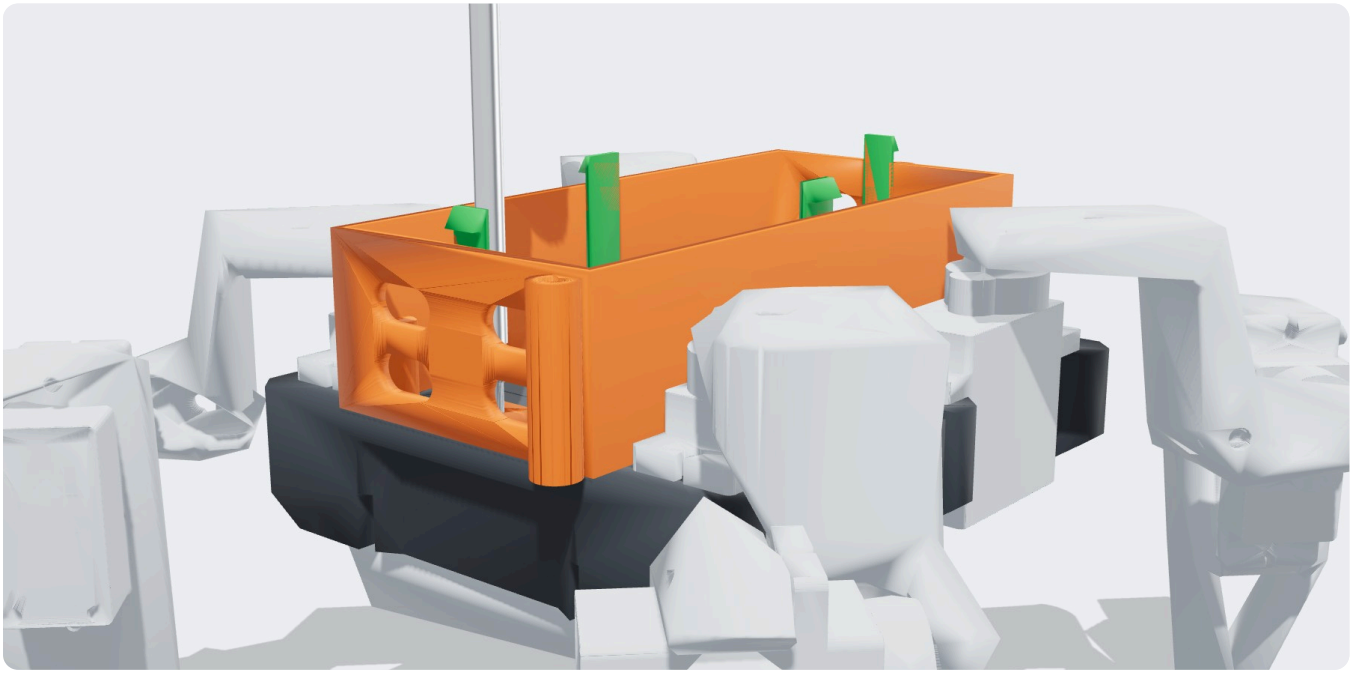
Build Sesame's base as in its build guide (`docs/build-guide/` in the repo): the servos in the internal frame, the legs on, the horns held by their M2.5 × 5 screws. Leave the bottom cover off.

With the robot upside down, push each servo's plug up through **its own slot** in the frame (one per servo, along the wire channels), then press the lead into the channel as Sesame's guide does. The leads come out inside the spine.



2 The spine onto the frame

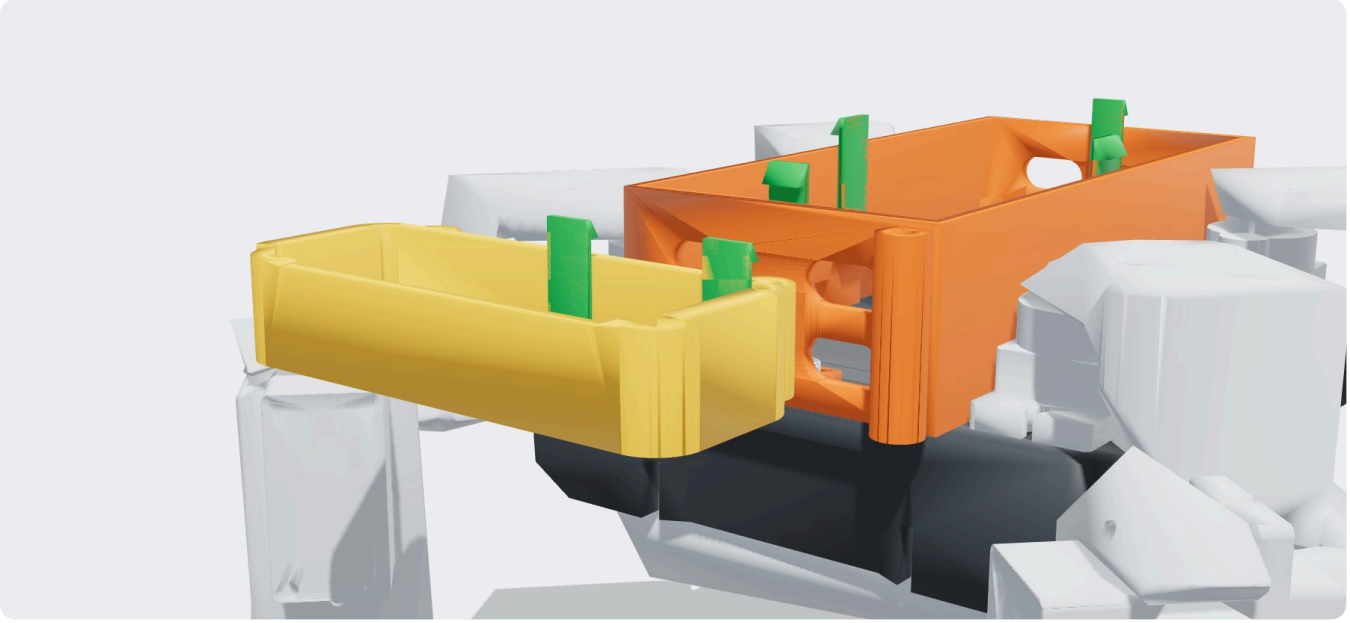
Set deck part 2a, the spine, on the frame. Screw it down with **4 screws up from below**, through the frame's corner holes, and **2 screws down from above**, through the pads at the foot of the spine's end walls. Reach them through the opening now, before anything goes on top. Then fit the battery, and the bottom cover with **2 screws**.



The second pair: down through the pads at the spine's ends. The four from below are already in.

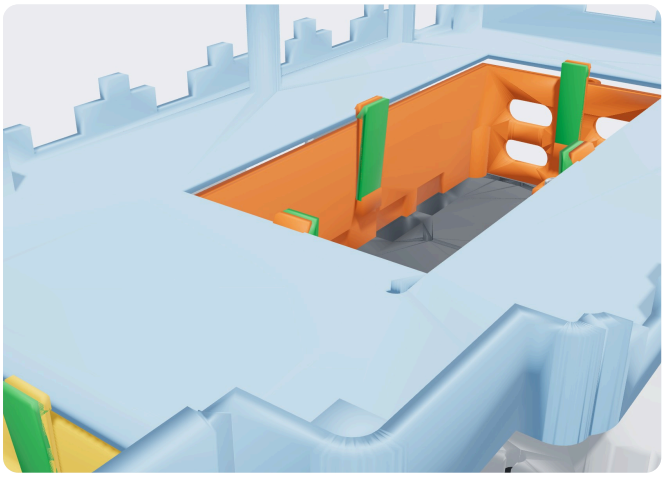
3 The head

Put the screen in deck part 2b, the head, face to the window, connector edge down, and set the head in front of the spine. Its two snap hooks stand behind the screen, on its back wall.

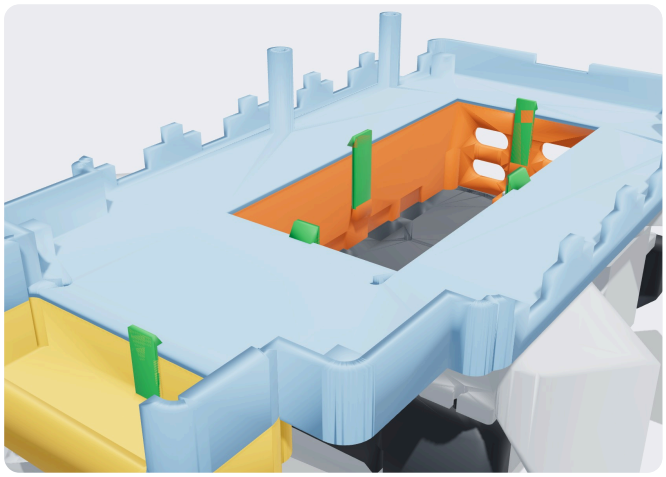


4 The plate snaps on

Lower deck part 1, the plate, floor-down over the spine and the head, with the servo leads up through its opening. Press it down evenly. The edge of the floor slides down each hook's ramp and bends it inward, then all **six click** back over the floor's edge. The keys on the spine and the head fit the recesses under the plate, so it only goes one way.



Pressing down: the floor rides the hook ramps and bends each hook in by about 1 mm.

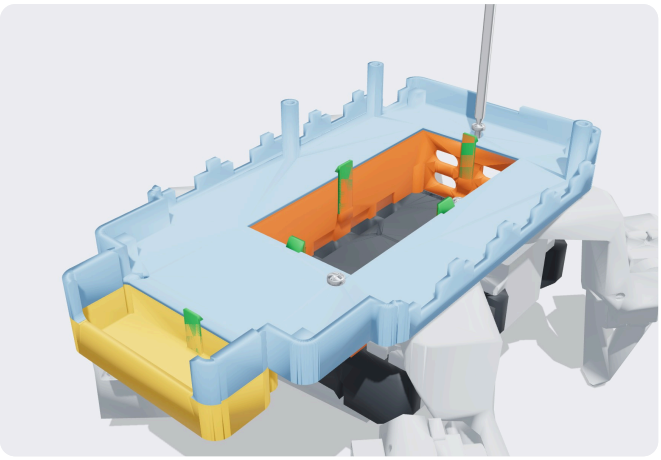


Clicked: all six over the floor's edge.

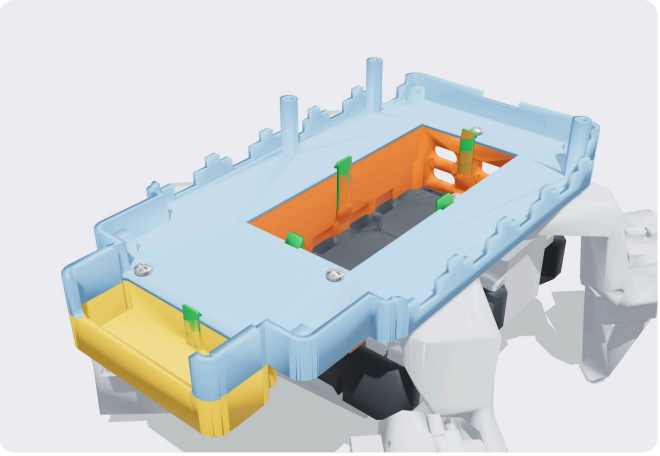
> The plate now holds by its hooks alone, enough to try it. To take it off, push each hook inward from above.

5 Screws for strength

Drive **4 screws** down through the plate's floor into the columns on the spine and the head: front right and back of the spine, left and right behind the head. The spine's front left column has no screw, because a screen ribbon lies there.



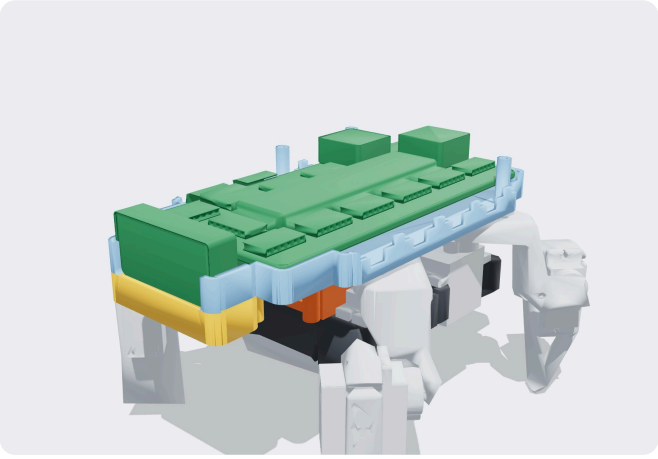
About 15 turns each, until the head seats.



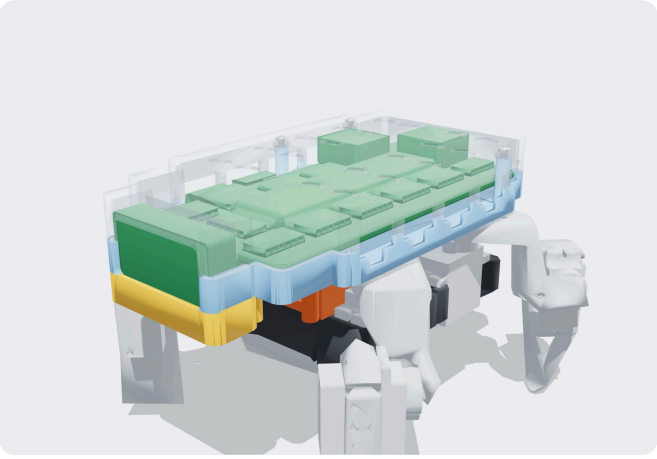
All four in.

6 Board, wiring, lid and bobbins

1. Set the board on the four posts. The USB-C module goes on **port 8**, the motion sensor on **port 4**, and the screen's cable to **ports 13 + 14**.
2. Wire the servos (page 6). Put each port's three jumpers in a **port jumper holder** and plug the holder in. Join each servo plug to its jumpers inside a **servo plug sleeve**.
3. Put the lid on with **3 screws**; their heads sink flush in its top.
4. Wind each leg's spare wire round a **wire bobbin** and pinch the ends in its slits. Drop its two pegs into the keyholes by that port on the lid and slide it 4 mm outward to lock.



The board on the plate's posts.



The lid on (shown clear, to see inside).

Wiring

PORT	LEG	LINE A (PIN 5)	LINE B (PIN 6)
2	front left	L1 hip	L3 knee
3	front right	R1 hip	R3 knee
10	back left	L2 hip	L4 knee
11	back right	R2 hip	R4 knee
4	motion sensor (icm40608)		
8	USB-C module		
13 + 14	screen, on its cable		

Each port is a 6-pin header: 1 +12 V, 2 +5 V, 3 +3.3 V, 4 GND, 5 line A, 6 line B. Run the signal and GND jumpers out through the slot at each port. Both sides have slots and lead notches (ports 2–6 on the right, 10–14 on the left), so a lead can leave on either side. Power the servos from the 5 V supply (red to +, brown to –), and join the supply's – to a port's GND.

> **+12 V sits next to +5 V, and which end of the header is pin 1 is not marked.** Measure before you wire. With the board on USB, the pin reading 3.3 V is pin 3; its neighbour that beeps to the USB-C shell is GND (pin 4). Never connect pin 1 to a servo.

Software

```
pip install atech
cd atech/firmware
atech validate .
atech upload .        # builds and flashes over USB
atech check           # which modules answered
```

About 2.5 s after it starts, the robot stands up and plays a show: wave, walk, turn, dance, crab, push-ups, bow. **Hold it off the table, or give it room, the first time.** Play any move from the laptop with `atech send spider_go walk`.

The files

Repository	`github.com/atechdotdev/ara-crab`
Branch	`main`
Everything for Atech-crab	`atech/`
Tutorial	`atech/README.md`
Print files	`atech/print/stl/ · stl_wires/ · gcode/`
This guide, the video, the 3D assembly	`atech/docs/`

Apache-2.0, like Sesame. Frame, servos, legs and moves: Sesame by Dorian Todd. Deck and lid: Atech Atelier's "Sesame walker", with wire openings, snap hooks and screws added.